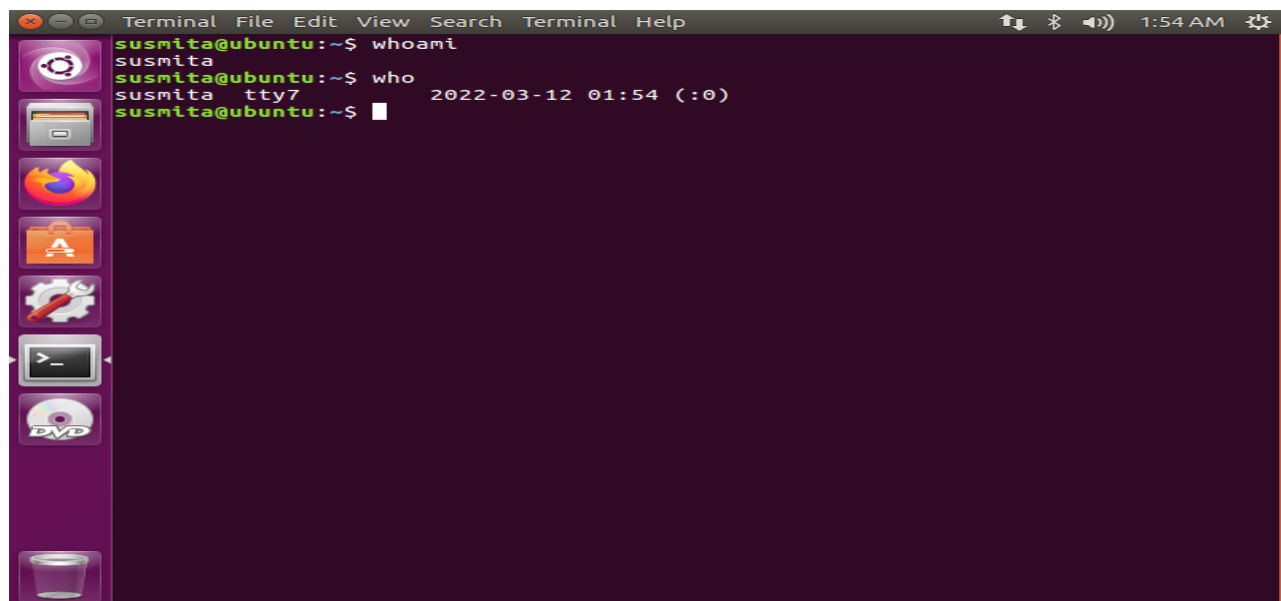
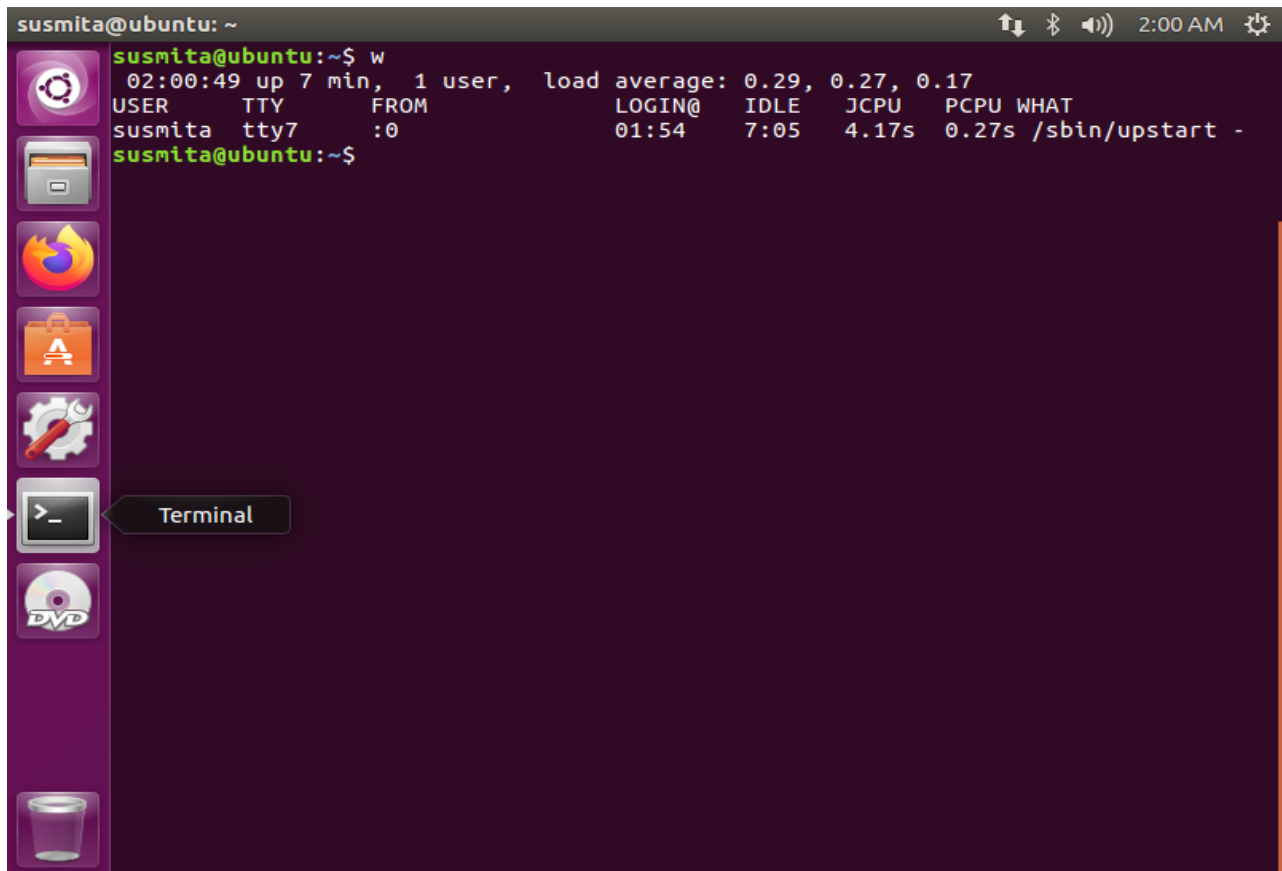

Let's have a look at some basic commands to find out more information about the system. They are:

- 1) **whoami** // It shows the username of the person who is logged into the system.

A screenshot of a Linux terminal window. The window has a title bar with 'Terminal' and menu options 'File', 'Edit', 'View', 'Search', 'Terminal', and 'Help'. The terminal content shows the user 'susmita' at the 'ubuntu' prompt. The first command is 'whoami', which outputs 'susmita'. The second command is 'who', which outputs 'susmita tty7 2022-03-12 01:54 (:0)'. The terminal has a dark purple background and a light purple sidebar on the left with various application icons.

```
Terminal File Edit View Search Terminal Help
susmita@ubuntu:~$ whoami
susmita
susmita@ubuntu:~$ who
susmita tty7 2022-03-12 01:54 (:0)
susmita@ubuntu:~$
```

- 2) **who** // It displays the list of users currently logged into the local system in detailed format. As we can observe it shows information such as the localhost/username of the system as well as the terminal on which the host is active and the date & time.
- 3) **w** // Shows who is logged in and what are they doing. Other system information such as login time, idle time, “tty” (tty command of terminal basically prints the file name of the terminal connected to standard input it is short of teletype), JCPU time (time used by all processes attached to the “tty” which does not include past background jobs but the currently executing background jobs), PCPU time (time used the current process named in the “what” field).

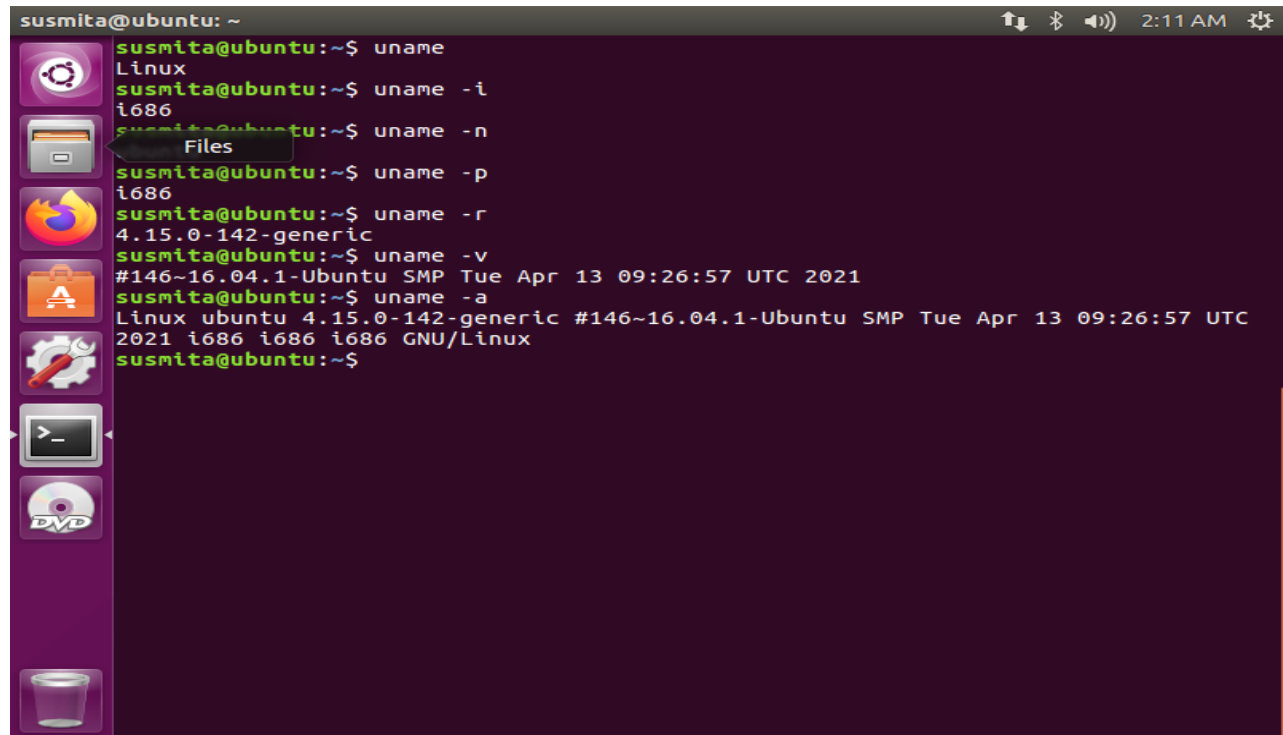


The screenshot shows a terminal window titled 'susmita@ubuntu: ~'. The terminal displays the output of the 'w' command, which provides system status and user information. The output includes the current time (02:00:49), system uptime (7 min), number of users (1), and load averages (0.29, 0.27, 0.17). Below this, a table lists the logged-in users with columns for USER, TTY, FROM, LOGIN@, IDLE, JCPU, PCPU, and WHAT. The table shows one user, 'susmita', logged in from 'tty7' at '01:54', with an idle time of '7:05', JCPU time of '4.17s', and PCPU time of '0.27s'. The 'WHAT' column shows the command '/sbin/upstart -'. The terminal window has a sidebar on the left with various application icons, and the 'Terminal' icon is highlighted. The top of the window shows system icons and the time '2:00 AM'.

```
susmita@ubuntu: ~  
susmita@ubuntu:~$ w  
02:00:49 up 7 min, 1 user, load average: 0.29, 0.27, 0.17  
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT  
susmita   tty7     :0               01:54    7:05   4.17s  0.27s  /sbin/upstart -  
susmita@ubuntu:~$
```

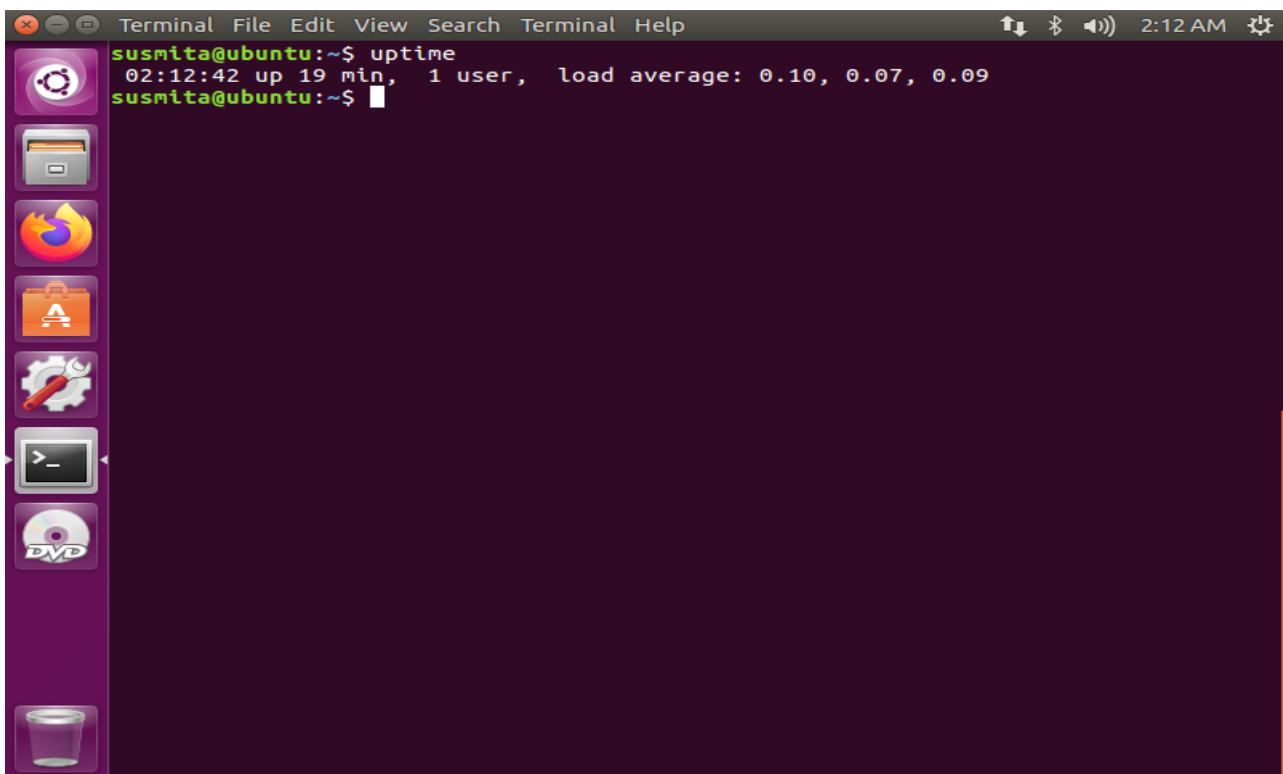
- 4) **uname**// gives away the machine characteristics. It has various associated options like,

- uname[-a]: all information related to machine
- uname[-i]: hardware implementation/platform
- uname[-n]: node name
- uname[-p]: processor
- uname[-r]: current kernel release
- uname[-v]: kernel version



A terminal window titled 'susmita@ubuntu: ~' with a top bar showing system icons and the time '2:11 AM'. The left sidebar contains icons for Dash, Home, Firefox, LibreOffice, Settings, Terminal, DVD, and Trash. The terminal output shows the following commands and results:

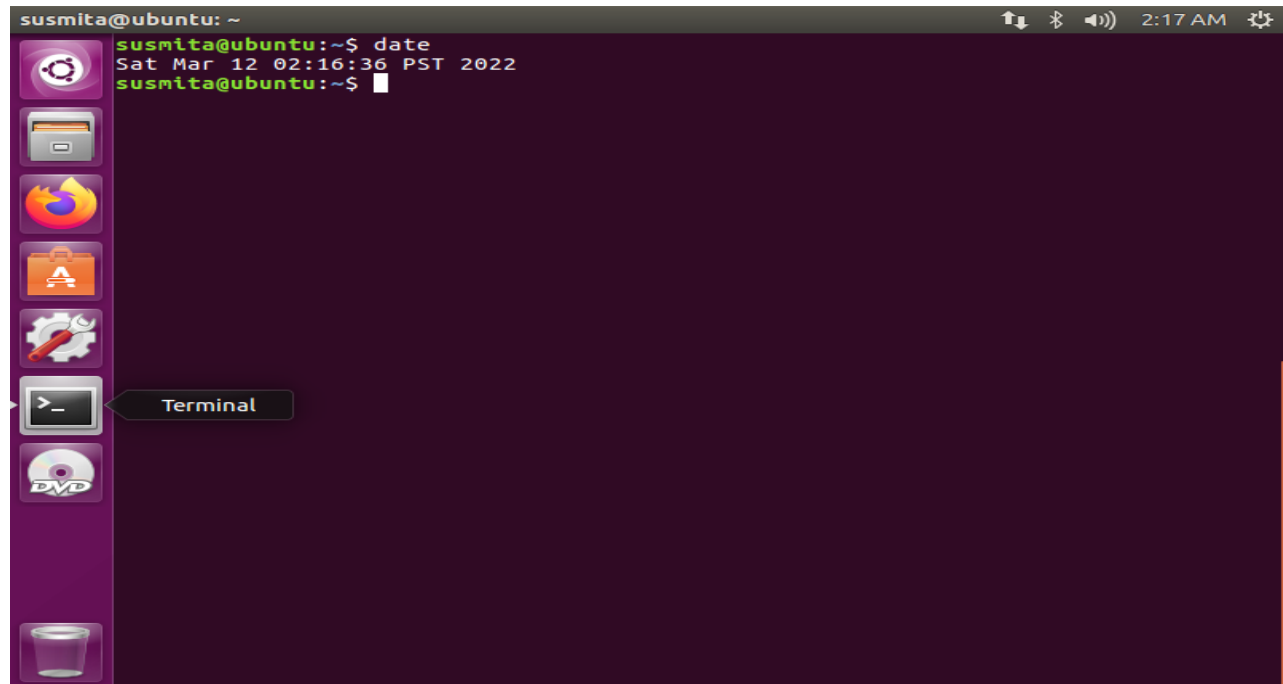
```
susmita@ubuntu:~$ uname
Linux
susmita@ubuntu:~$ uname -i
i686
susmita@ubuntu:~$ uname -n
Files
susmita@ubuntu:~$ uname -p
i686
susmita@ubuntu:~$ uname -r
4.15.0-142-generic
susmita@ubuntu:~$ uname -v
#146~16.04.1-Ubuntu SMP Tue Apr 13 09:26:57 UTC 2021
susmita@ubuntu:~$ uname -a
Linux ubuntu 4.15.0-142-generic #146~16.04.1-Ubuntu SMP Tue Apr 13 09:26:57 UTC 2021 i686 i686 i686 GNU/Linux
susmita@ubuntu:~$
```



A terminal window titled 'Terminal' with a top bar showing system icons and the time '2:12 AM'. The left sidebar is identical to the first image. The terminal output shows the following command and result:

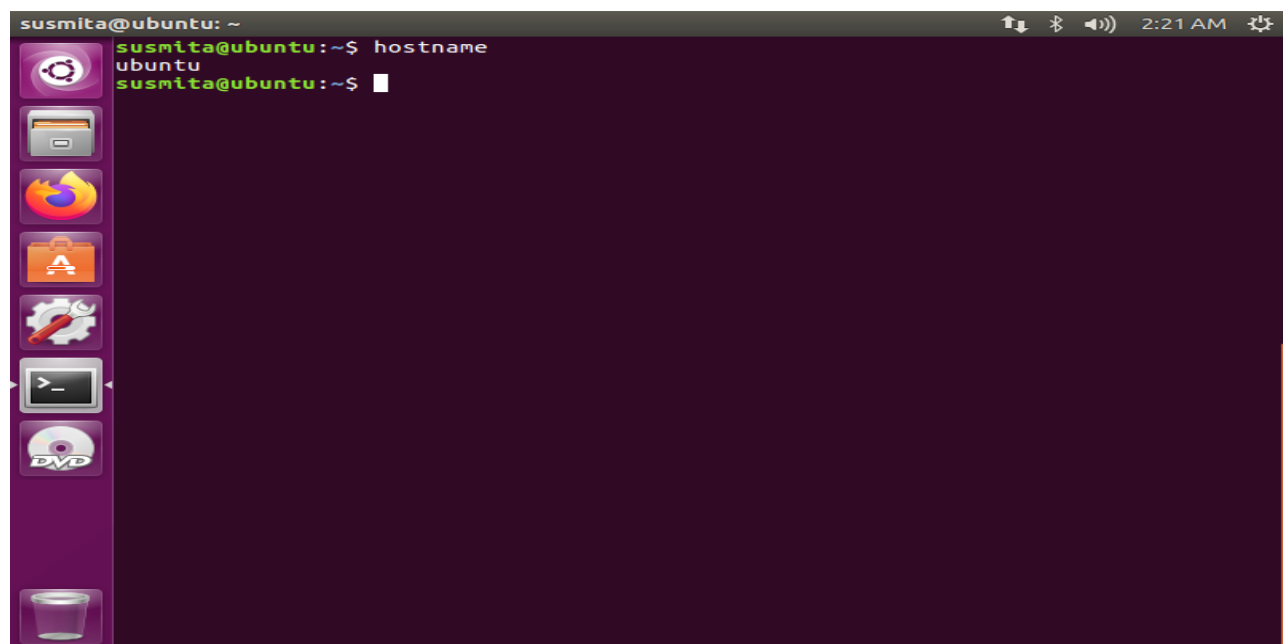
```
susmita@ubuntu:~$ uptime
02:12:42 up 19 min,  1 user,  load average: 0.10, 0.07, 0.09
susmita@ubuntu:~$
```

- 5) **uptime** // It tells how long the system has been running.
- 6) **date** // It is used to display or set the date. **Set the date requires “root” access.**

A terminal window titled 'susmita@ubuntu: ~' with a dark purple background. The left sidebar contains icons for Dash, Home, Firefox, Files, Applications, Settings, Terminal, DVD, and Trash. The Terminal icon is highlighted with a tooltip that says 'Terminal'. The terminal text shows the user running the 'date' command, which outputs 'Sat Mar 12 02:16:36 PST 2022'.

```
susmita@ubuntu: ~  
susmita@ubuntu:~$ date  
Sat Mar 12 02:16:36 PST 2022  
susmita@ubuntu:~$
```

7) **hostname** // command displays the name of the computer

A terminal window titled 'susmita@ubuntu: ~' with a dark purple background. The left sidebar contains icons for Dash, Home, Firefox, Files, Applications, Settings, Terminal, DVD, and Trash. The Terminal icon is highlighted with a tooltip that says 'Terminal'. The terminal text shows the user running the 'hostname' command, which outputs 'ubuntu'.

```
susmita@ubuntu: ~  
susmita@ubuntu:~$ hostname  
ubuntu  
susmita@ubuntu:~$
```

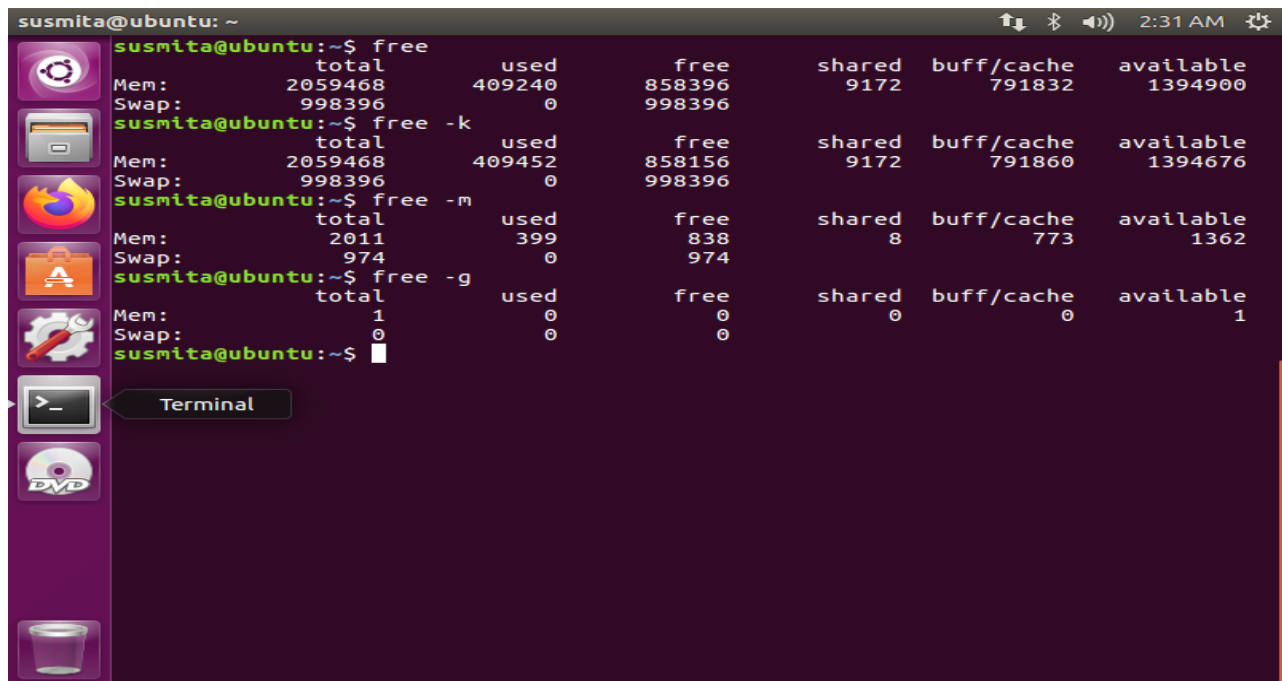
8) **cal** // It displays the calendar & **cal -h** displays the entire year's calendar. Other combinations for displaying the calendar include the following:

cal -j : It displays the day according to the Julian calendar (days numbered 1 to 365, starting from January 1 onwards)

```
susmita@ubuntu: ~  
susmita@ubuntu:~$ cal  
Search your computer  
Su Mo Tu We Th Fr Sa  
        1  2  3  4  5  
 6  7  8  9 10 11 12  
13 14 15 16 17 18 19  
20 21 22 23 24 25 26  
27 28 29 30 31  
susmita@ubuntu:~$
```

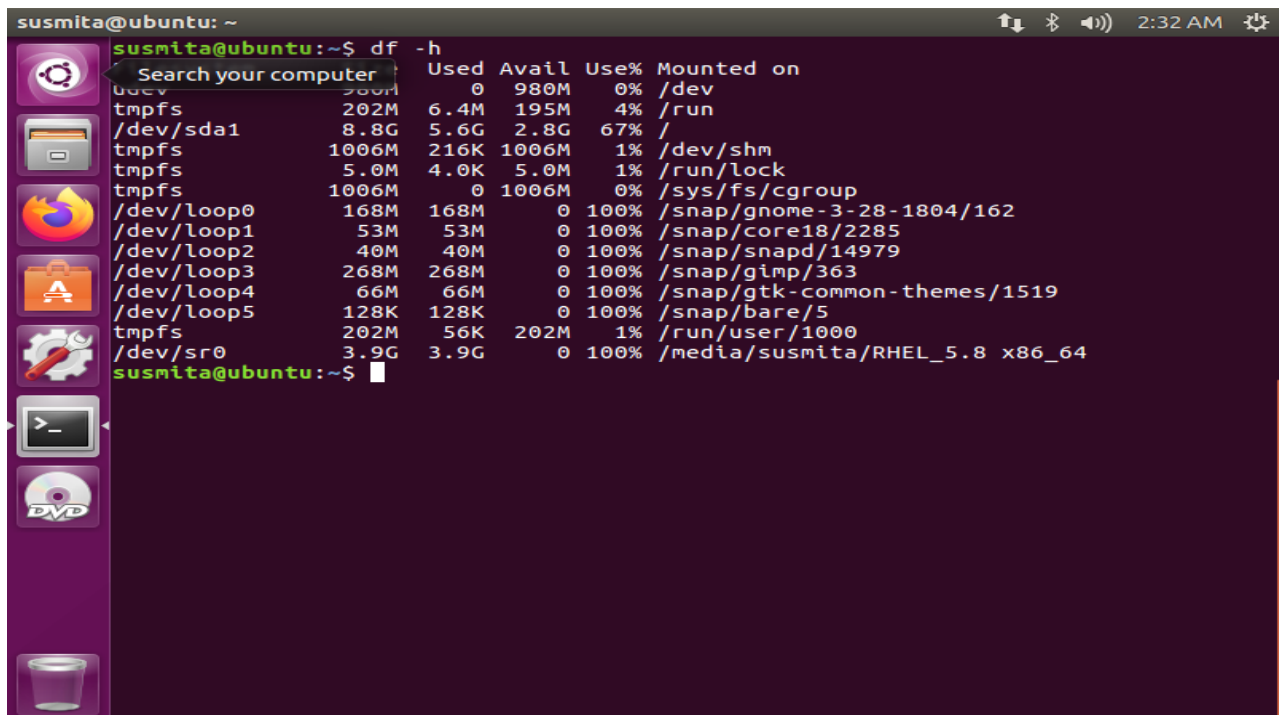
```
Terminal File Edit View Search Terminal Help  
susmita@ubuntu:~$ cal -hyy  
2022  
January February  
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa  
        1  2  3  4  5  6  7  8  37 38 39 40 41 42 43  
 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 58 59  
16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31  
30 31  
March April  
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa  
        60 61 62 63 64 91 92  
65 66 67 68 69 70 71 93 94 95 96 97 98 99  
72 73 74 75 76 77 78 100 101 102 103 104 105 106  
79 80 81 82 83 84 85 107 108 109 110 111 112 113  
86 87 88 89 90 114 115 116 117 118 119 120  
May June  
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa  
121 122 123 124 125 126 127 152 153 154 155  
128 129 130 131 132 133 134 156 157 158 159 160 161 162  
135 136 137 138 139 140 141 163 164 165 166 167 168 169  
142 143 144 145 146 147 148 170 171 172 173 174 175 176  
149 150 151 177 178 179 180 181  
July August  
Su Mo Tu We Th Fr Sa Su Mo Tu We Th Fr Sa  
        182 183 213 214 215 216 217 218  
184 185 186 187 188 189 190 219 220 221 222 223 224 225
```

- 9) **free** // It displays the amount of free and used memory in the system.
free -k // output in Kilobytes
free -g // output in Gigabytes
free -m // output in Megabytes



```
susmita@ubuntu: ~  
susmita@ubuntu:~$ free  
              total        used        free      shared  buff/cache   available  
Mem:           2059468      409240        858396         9172       791832      1394900  
Swap:           998396           0        998396  
susmita@ubuntu:~$ free -k  
              total        used        free      shared  buff/cache   available  
Mem:           2059468      409452        858156         9172       791860      1394676  
Swap:           998396           0        998396  
susmita@ubuntu:~$ free -m  
              total        used        free      shared  buff/cache   available  
Mem:             2011          399           838           8           773          1362  
Swap:             974           0           974  
susmita@ubuntu:~$ free -g  
              total        used        free      shared  buff/cache   available  
Mem:              1           0           0           0           0           1  
Swap:              0           0           0  
susmita@ubuntu:~$
```

- 10) **df -h** // Displays information about the total space & available space on a filesystem. Along with this information it also lets us know about the local disk partitions like



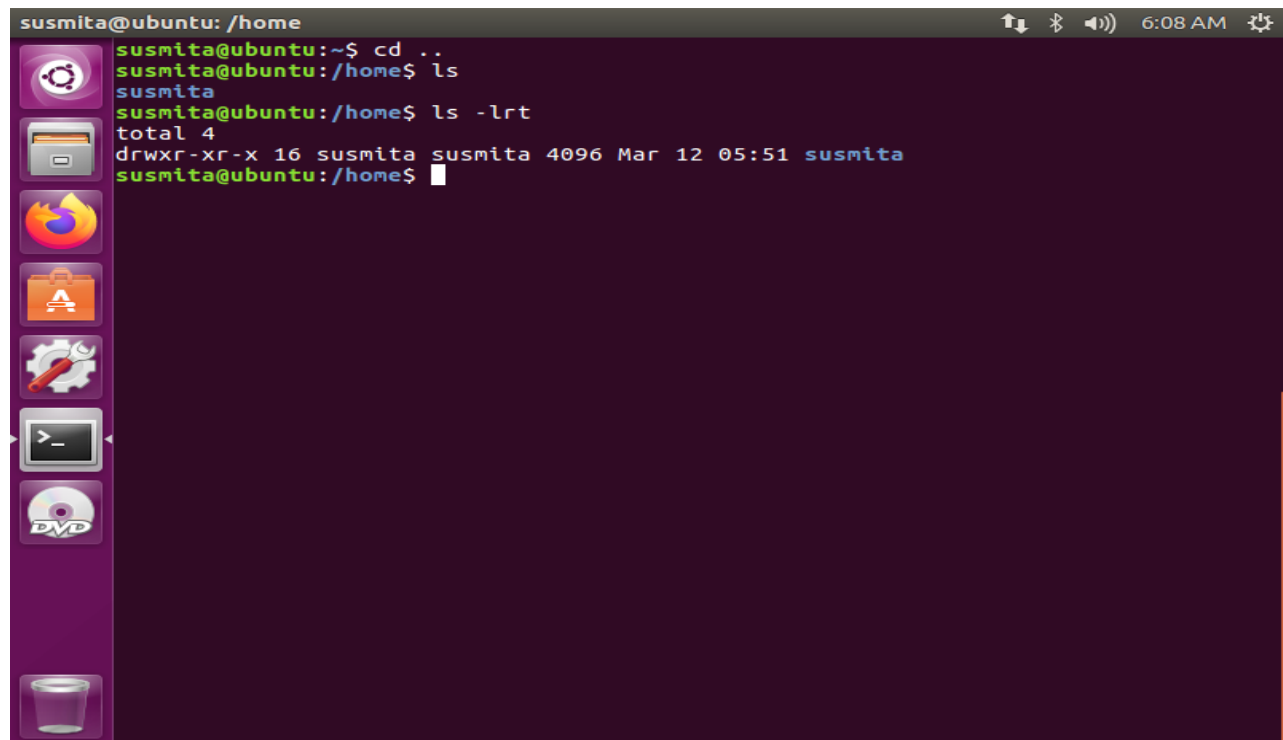
```
susmita@ubuntu: ~  
susmita@ubuntu:~$ df -h  
Filesystem      Used Avail Use% Mounted on  
udev            0 980M  0% /dev  
tmpfs            202M 6.4M  4% /run  
/dev/sda1       8.8G 5.6G  67% /  
tmpfs           1006M 216K 100% /dev/shm  
tmpfs            5.0M 4.0K  1% /run/lock  
tmpfs           1006M 0 100% /sys/fs/cgroup  
/dev/loop0      168M 168M  0% /snap/gnome-3-28-1804/162  
/dev/loop1       53M  53M  0% /snap/core18/2285  
/dev/loop2       40M  40M  0% /snap/snapd/14979  
/dev/loop3      268M 268M  0% /snap/gimp/363  
/dev/loop4       66M  66M  0% /snap/gtk-common-themes/1519  
/dev/loop5      128K 128K  0% /snap/bare/5  
tmpfs            202M 56K  1% /run/user/1000  
/dev/sr0        3.9G 3.9G  0% /media/susmita/RHEL_5.8_x86_64  
susmita@ubuntu:~$
```

We, have verified the basic commands to know more about the installed Linux system.

Demonstration of Linux commands (which map standard Graphical UIs):

Let's have a look at the commands which can be used to create directories and files. We will create folders/directories and also have a look at the commands which can be used to create text files, vi Editor is the most common editor used to create text files. During the course of executing these commands, we will also understand more about changing directories or paths in our Linux system.

- 11) **mkdir** // It is used to create directory/ directories. We, use the “ls” command to list the directories and files in our system. By default, you are in the “home” directory of your machine which is generally listed under “username” of the system.

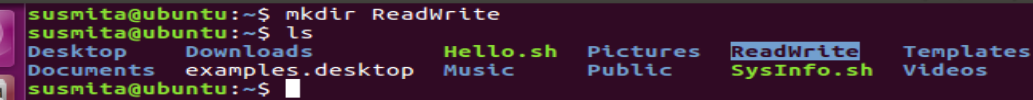
A screenshot of a Linux terminal window with a dark purple background. The window title bar shows 'susmita@ubuntu: /home' and system icons on the right. On the left side, there is a vertical dock with icons for Dash, Home Folder, Firefox, Dash to Dock, Software Center, Settings, Terminal, DVD, and Trash. The terminal text shows the following sequence of commands and output:

```
susmita@ubuntu:~$ cd ..
susmita@ubuntu:/home$ ls
susmita
susmita@ubuntu:/home$ ls -lrt
total 4
drwxr-xr-x 16 susmita susmita 4096 Mar 12 05:51 susmita
susmita@ubuntu:/home$
```

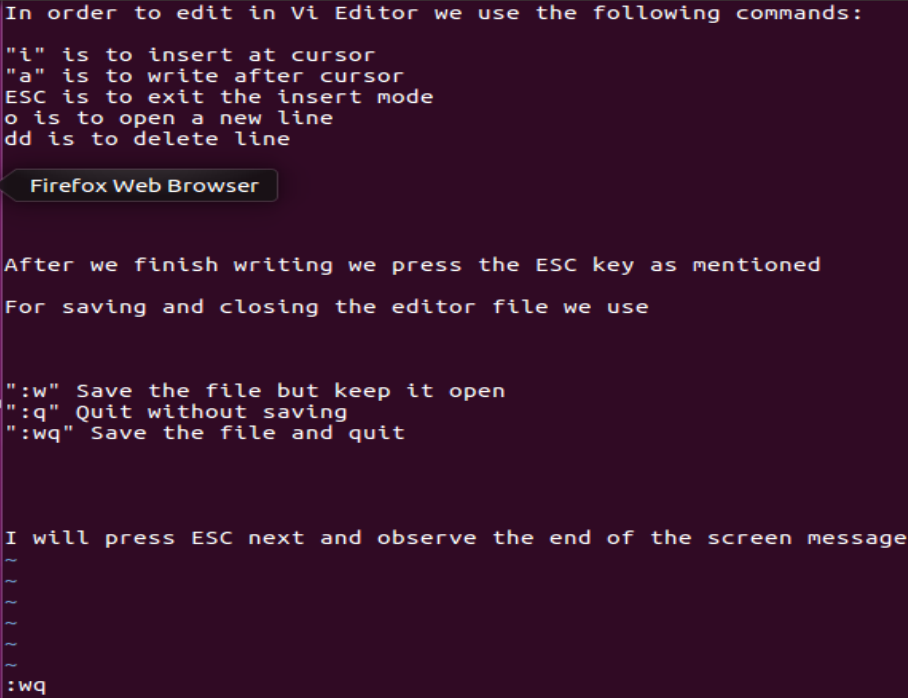
You can use the **cd** command, that is **change the directory** to move to your **default home directory** which stores all the user created directories by default as discussed in the Linux Filesystem structure.

You can verify by checking the “**home**” directory in your system after using the “**mkdir**” command if a directory has been created in your system. We are creating our directory by the name “**ReadWrite**” as we wish to explore reading/writing into files.

We can also verify its listing either through commands or changing the path to be sure of it's creation.



After creating the directory, we use the **vi Editor** to start creating text files. We will use **Editor.txt** file to list down the commands/instructions to create and edit files in vi Editor. Following next is the commands for editing text files with vi Editor.



To check the demonstration for cat and vi Editor commands we have,


```
susmita@ubuntu: ~/ReadWrite
susmita@ubuntu:~/ReadWrite$ vi Editor.txt
susmita@ubuntu:~/ReadWrite$ cat Editor.txt
In order to edit in Vi Editor we use the following commands:

"i" is to insert at cursor
"a" is to write after cursor
ESC is to exit the insert mode
o is to open a new line
dd is to delete line

After we finish writing we press the ESC key as mentioned
System Settings
For saving and closing the editor file we use

":w" Save the file but keep it open
":q" Quit without saving
":wq" Save the file and quit

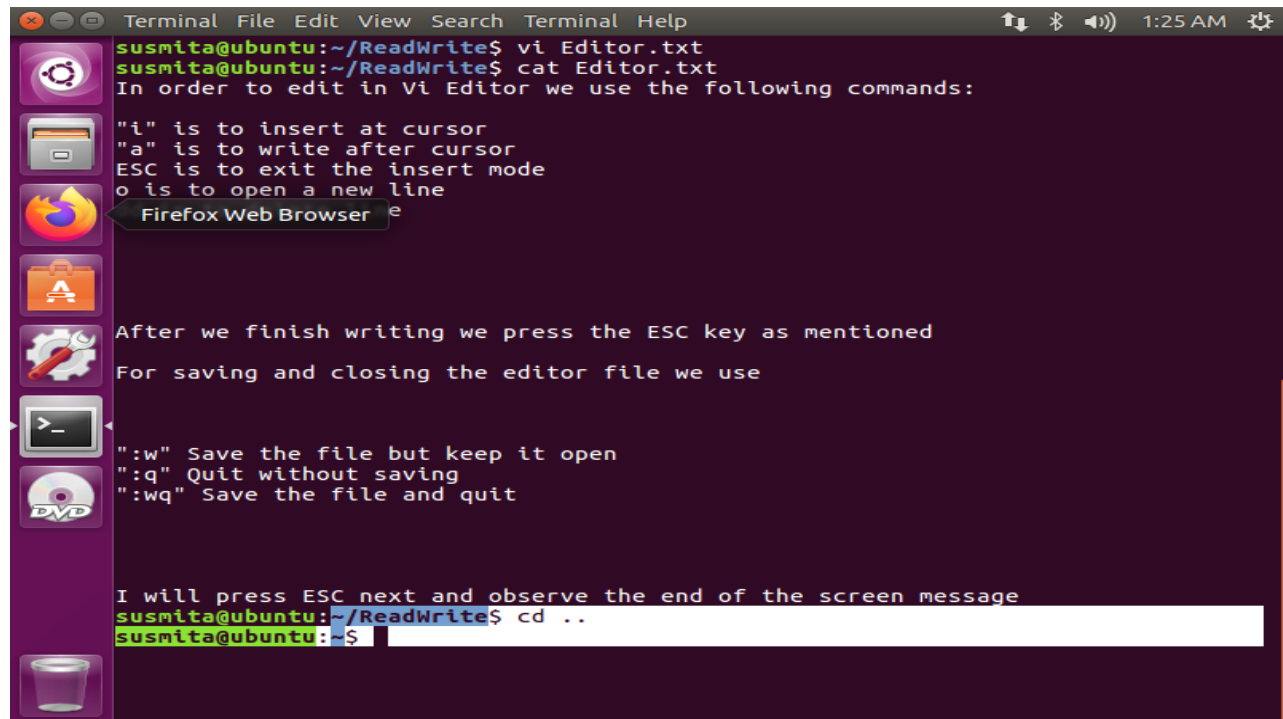
I will press ESC next and observe the end of the screen message
susmita@ubuntu:~/ReadWrite$
```

The other commands which we discussed in context to files and directories are as follows:

12) **cd <Directory-name>**// change the directory as required by user

```
susmita@ubuntu: ~/ReadWrite
susmita@ubuntu:~$ ls
Desktop    Downloads  Hello.sh  Pictures  ReadWrite  Templates
Documents  examples.desktop  Music     Public    SysInfo.sh  Videos
susmita@ubuntu:~$ cd ReadWrite
susmita@ubuntu:~/ReadWrite$ ls
Demo.txt  Editor.txt
susmita@ubuntu:~/ReadWrite$
```

13) **cd ..** // is used to go back to the parent directory



The image shows a terminal window with a purple background and a sidebar on the left containing icons for various applications. The terminal output is as follows:

```
susmita@ubuntu:~/ReadWrite$ vi Editor.txt
susmita@ubuntu:~/ReadWrite$ cat Editor.txt
In order to edit in Vi Editor we use the following commands:

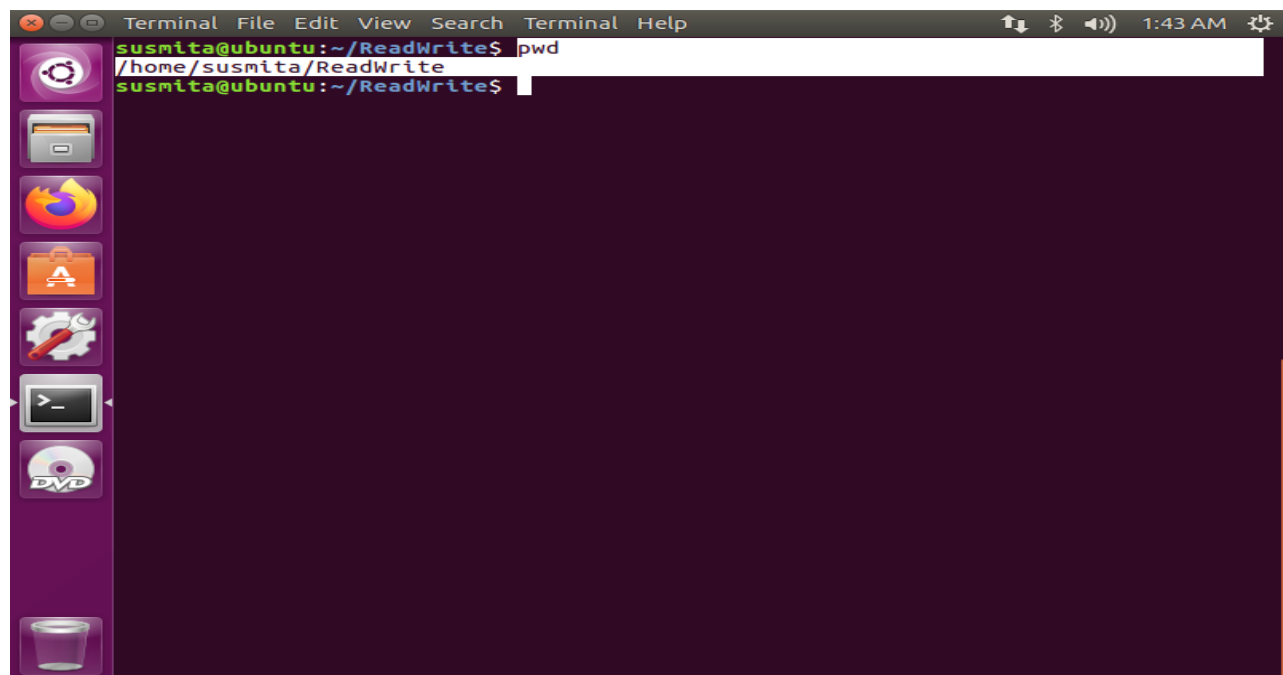
"i" is to insert at cursor
"a" is to write after cursor
ESC is to exit the insert mode
o is to open a new line

After we finish writing we press the ESC key as mentioned
For saving and closing the editor file we use

":w" Save the file but keep it open
":q" Quit without saving
":wq" Save the file and quit

I will press ESC next and observe the end of the screen message
susmita@ubuntu:~/ReadWrite$ cd ..
susmita@ubuntu:~$
```

We use the “**pwd**” command to list the current/present working directory.



The image shows a terminal window with a purple background and a sidebar on the left containing icons for various applications. The terminal output is as follows:

```
susmita@ubuntu:~/ReadWrite$ pwd
/home/susmita/ReadWrite
susmita@ubuntu:~/ReadWrite$
```

14) **pwd** // present working directory

We can also, use **cat** and **touch** commands in order to create text files. We will see their application in context to the command **cp**, that is the **copy** command.

```
susmita@ubuntu: ~/WriteReadExecute
susmita@ubuntu:~/WriteReadExecute$ cat >Demo.txt
Search your computer sert into a text file through the cat command.
[5]+ Stopped cat > Demo.txt
susmita@ubuntu:~/WriteReadExecute$ cat Demo.txt
We use the ">" to insert into a text file through the cat command.
susmita@ubuntu:~/WriteReadExecute$
```

15) Creating/Reading with the help of **cat** command are different in syntax.

cat > Filename.txt // to create a file

cat Filename.txt // to read a file

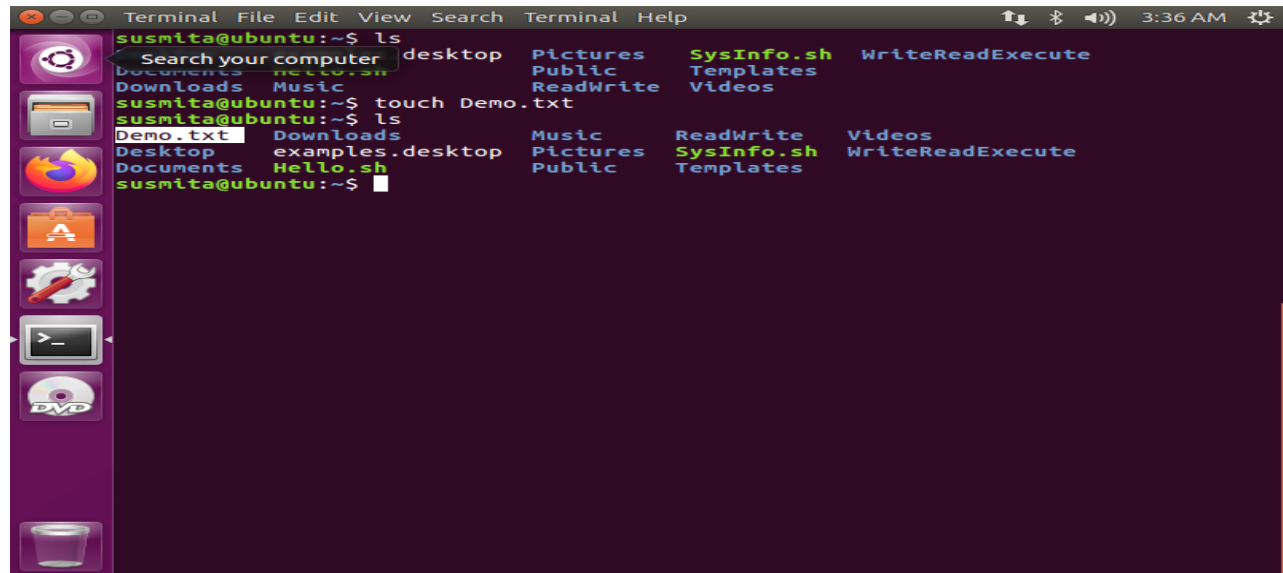
```
susmita@ubuntu: ~/WriteReadExecute
susmita@ubuntu:~/WriteReadExecute$ cat >Demo.txt
Search your computer sert into a text file through the cat command.
[5]+ Stopped cat > Demo.txt
susmita@ubuntu:~/WriteReadExecute$ cat Demo.txt
We use the ">" to insert into a text file through the cat command.
susmita@ubuntu:~/WriteReadExecute$ touch Example.txt
susmita@ubuntu:~/WriteReadExecute$ ls
Demo.txt Example.txt
susmita@ubuntu:~/WriteReadExecute$
```

```
susmita@ubuntu: ~/WriteReadExecute
susmita@ubuntu:~/WriteReadExecute$ cat >Demo.txt
Search your computer insert into a text file through the cat command.
[5]+  Stopped                  cat > Demo.txt
susmita@ubuntu:~/WriteReadExecute$ cat Demo.txt
We use the ">" to insert into a text file through the cat command.
susmita@ubuntu:~/WriteReadExecute$ touch Example.txt
susmita@ubuntu:~/WriteReadExecute$ ls
Demo.txt  Example.txt
susmita@ubuntu:~/WriteReadExecute$
```

16) **cp** : To copy the contents from one file to another we use the “**cp**” command.

```
susmita@ubuntu: ~/WriteReadExecute
susmita@ubuntu:~/WriteReadExecute$ cat >Demo.txt
We use the ">" to insert into a text file through the cat command.
^Z
[5]+  Stopped                  cat > Demo.txt
susmita@ubuntu:~/WriteReadExecute$ cat Demo.txt
We use the ">" to insert into a text file through the cat command.
susmita@ubuntu:~/WriteReadExecute$ touch Example.txt
susmita@ubuntu:~/WriteReadExecute$ ls
Demo.txt  Example.txt
susmita@ubuntu:~/WriteReadExecute$ cp Demo.txt Example.txt
susmita@ubuntu:~/WriteReadExecute$ cat Example.txt
We use the ">" to insert into a text file through the cat command.
susmita@ubuntu:~/WriteReadExecute$
```

17) **rm/ rmdir** : In order to delete files we use the “**rm**” command and for directories the “**rmdir**” command. In case a directory cannot be deleted directly (which happens if it’s not empty) we use the “**rm -rf**” command (forceful deletion of the directory).

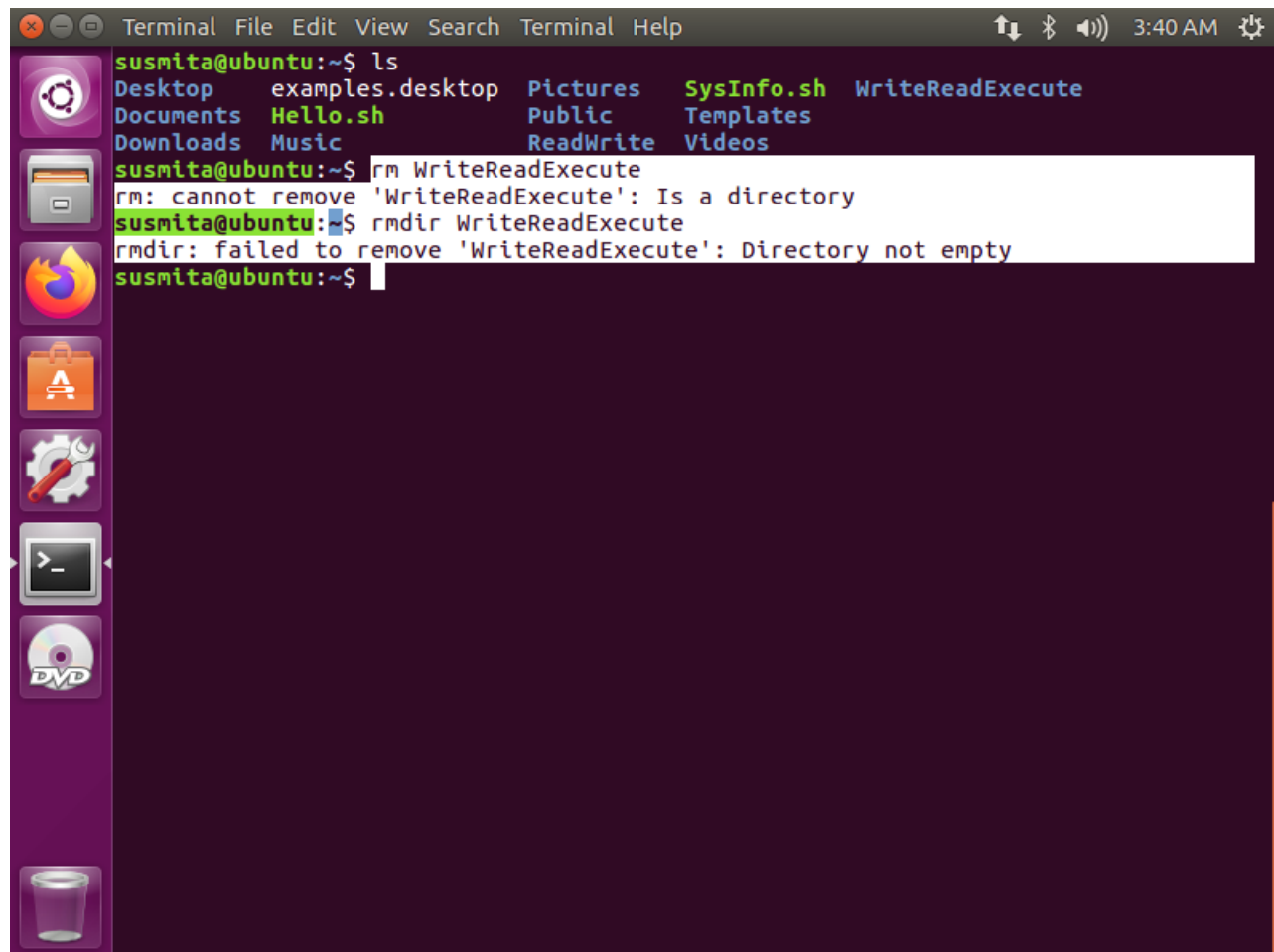


A terminal window titled 'Terminal' with a menu bar (File, Edit, View, Search, Terminal, Help) and a status bar (3:36 AM). The user 'susmita@ubuntu' is at the prompt '~\$'. They run 'ls' and see a list of files and directories: Desktop, Pictures, SysInfo.sh, WriteReadExecute, Documents, Hello.sh, Public, Templates, Downloads, Music, ReadWrite, Videos, Demo.txt, examples.desktop. They then run 'touch Demo.txt'.

```
susmita@ubuntu:~$ ls
Desktop  Pictures  SysInfo.sh  WriteReadExecute
Documents Hello.sh   Public      Templates
Downloads Music      ReadWrite   Videos
Demo.txt examples.desktop

susmita@ubuntu:~$ touch Demo.txt
susmita@ubuntu:~$
```

In order to **delete/remove** a directory,



A terminal window titled 'Terminal' with a menu bar (File, Edit, View, Search, Terminal, Help) and a status bar (3:40 AM). The user 'susmita@ubuntu' is at the prompt '~\$'. They run 'ls' and see the same list of files and directories as before. They then run 'rm WriteReadExecute' and get an error: 'rm: cannot remove 'WriteReadExecute': Is a directory'. They then run 'rmdir WriteReadExecute' and get another error: 'rmdir: failed to remove 'WriteReadExecute': Directory not empty'.

```
susmita@ubuntu:~$ ls
Desktop  examples.desktop  Pictures  SysInfo.sh  WriteReadExecute
Documents Hello.sh          Public    Templates
Downloads Music              ReadWrite Videos

susmita@ubuntu:~$ rm WriteReadExecute
rm: cannot remove 'WriteReadExecute': Is a directory

susmita@ubuntu:~$ rmdir WriteReadExecute
rmdir: failed to remove 'WriteReadExecute': Directory not empty

susmita@ubuntu:~$
```

Hence, we use the forceful deletion command to **remove/delete** the directory.

```
susmita@ubuntu: ~  
susmita@ubuntu:~$ ls  
Desktop  examples.desktop  Pictures  SysInfo.sh  WriteReadExecute  
Documents Hello.sh          Public   Templates  
Downloads Music           ReadWrite Videos  
susmita@ubuntu:~$ rmdir ReadWrite  
rmdir: failed to remove 'ReadWrite': Directory not empty  
susmita@ubuntu:~$ rm -rf ReadWrite  
susmita@ubuntu:~$ ls  
Desktop  examples.desktop  Hello.sh  Pictures  SysInfo.sh  Videos  
Documents examples.desktop  Music     Public   Templates  WriteReadExecute  
susmita@ubuntu:~$
```

18) To **move** a file from one directory to another directory/subdirectory we use the “**mv**” command.

```
Terminal File Edit View Search Terminal Help  
susmita@ubuntu:~$ ls  
Desktop  examples.desktop  Pictures  SysInfo.sh  WriteReadExecute  
Documents Hello.sh          Public   Templates  
Downloads Music           ReadWrite Videos  
susmita@ubuntu:~$ rmdir ReadWrite  
rmdir: failed to remove 'ReadWrite': Directory not empty  
susmita@ubuntu:~$ rm -rf ReadWrite  
susmita@ubuntu:~$ ls  
Desktop  Downloads  Hello.sh  Pictures  SysInfo.sh  Videos  
Documents examples.desktop  Music     Public   Templates  WriteReadExecute  
susmita@ubuntu:~$ cd WriteReadExecute  
susmita@ubuntu:~/WriteReadExecute$ ls  
Demo.txt  Example.txt  
susmita@ubuntu:~/WriteReadExecute$ mkdir Directories  
susmita@ubuntu:~/WriteReadExecute$ mv Example.txt Directories  
susmita@ubuntu:~/WriteReadExecute$ cd Directories  
susmita@ubuntu:~/WriteReadExecute/Directories$ ls  
Example.txt  
susmita@ubuntu:~/WriteReadExecute/Directories$
```

Hence, we have demonstrated the commands for finding the basic information about the system as well as executed the commands for creating directories and files which map standard Graphical User Interfaces (GUIs).